

Anticipated Jury Questions

Methodology & Simulation (Section 3)

Deniz Ege Memetoğlu • FENG 498 • assumes the jury has the report and the presentation in front of them

Model design

Q. Is this a trace replay of the logs, or a real model?

Distribution-driven, not a replay. I fit distributions (inter-batch gap, batch size, kit composition, service times) from the four-month ZWM92 dispatch log, 167,784 rows and 40,804 reconstructed kit orders, and the model samples from them. It does not play history back. That is deliberate: because demand is generated and not fixed, I can move materials to new locations and re-run the same warehouse under a different policy, which a replay could never do. It is built in Python with SimPy.

↔ Report: Methodology, Simulation Model (Modelling Approach, Order Generation). Slide: the methodology flowchart and the distribution-driven logic slide.

Q. Why discrete-event simulation, and why SimPy?

The bottleneck is queueing for shared resources: operators waiting on a seven-truck reach-truck fleet, plus the cascade where one order holds its slot while it waits and delays the next. Static or analytic models assume average flow and miss that contention. SimPy is a mature, transparent discrete-event library; every resource, operators, reach trucks, Kardex units and pallet-position locks, is an explicit, auditable object.

↔ Report: Methodology, Simulation Model. Slide: the SimPy resource-model figure.

Inputs and calibration

Q. Where do the pick and handling times come from?

From a video time-motion study of the F400 kitting line: 2,319 timed micro-events over 296.6 observed minutes. Service times are lognormal, with the spread set by the measured coefficient of variation through $\sigma = \sqrt{\ln(1 + CV^2)}$. The means are operator pick 0.113 min, reach-truck pick-and-place 0.110, manual-band penalty 0.102, and Kardex pick 0.113; the sigmas are 1.245, 1.047 and 1.279.

↔ Report: Methodology, Data Collection (Time and Motion Study) and Simulation Model (Travel, Picking and Service Times). Slide: the sigma formula on the logic slide.

Q. How did you validate the Kardex timing? There was no Kardex video.

Honest answer: there was no Kardex-specific footage, so the Kardex pick mean and CV mirror the F400 operator pick, with a 0.4-minute carousel turn as a book value. We disclose this in the assumption register. It cannot bias the comparison, because Kardex routing is identical across all six policies, so any timing error cancels out of the ranking.

↔ Report: Appendix, Model Assumption Register.

Q. F400 timing is extrapolated to nine families. What if F400 is unrepresentative?

F400 is about 29% of the dispatch rows, and its timing is applied to the other eight families. We bound that risk with a one-at-a-time sensitivity sweep of plus or minus twenty percent on every timing input. Line-aware stays the best policy under every swing; lead time moves proportionally, but the ranking never flips. Per-line video studies are listed as future work.

↔ Report: Results, Sensitivity Analysis (tornado figure).

Q. Why model batch arrivals, and does the rate match reality?

Because the real log arrives in bursts: 3,885 same-timestamp batches, mean 4.68 kits per batch, with an inter-batch gap mean of 5.36 minutes and a CV of 2.04, so it is empirical, not Poisson. Across the full twenty-replication set the model produces about 400 orders a day, within 1.1% of the real 396-per-day average.

↔ Report: Methodology, Simulation Model (Order Generation); Results, Verification and Validation Results (daily-volume calibration).

Experiment design

Q. Why twenty replications, and why the same seeds across policies?

Each policy runs twenty replications; replication i uses seed $42 + i$, with separate random streams for arrivals and for service times. Common random numbers mean every policy faces exactly the same demand, so the differences are the policy, not luck. The 95% confidence intervals, using the t-distribution with 19 degrees of freedom, are narrow and do not overlap between Line-aware and the baselines, which is why twenty runs are enough.

↔ Report: Methodology, Experimental Design and Statistical Analysis (replication analysis). Slide: the statistics slide.

Q. Why six policies, not only your proposed one?

To place the proposal against a fair spread: the real SAP placement as the operational baseline, a simple capacity heuristic as a control, two literature-based ABC variants, a pure travel-distance optimizer, and our Line-aware policy. Without those references, “better” would mean nothing. The policy design itself is Nehir’s section, but the experimental frame is shared.

↔ Report: Methodology, Storage Assignment Policies.

Verification and validation

Q. Your chi-square is highly significant. Doesn’t that mean the model is wrong?

No, and this is the key point. The pooled chi-square with 7 degrees of freedom is 535, p below 0.001, but Cramer’s V is only 0.22, a small-to-moderate effect. It is significant precisely because the model uses its own routing logic, not a bin-for-bin replay; an exact match would only happen if we replayed history, which would defeat the purpose. We read the effect size, and we treat it as an internal-consistency check.

↔ Report: Results, Verification and Validation Results. Slide: the validation slide with the chi-square and Cramer’s V formula.

Q. You fitted and validated on the same ZWM92 data. No holdout?

Correct, and we disclose it. SAP never logged waiting times or reach-truck calls, so there was no independent observation period to hold out. The independent grounding is the F400 video for timing,

the daily-volume calibration to within about one percent, and a 3D route audit in which no agent crosses a rack body. We label these as internal-consistency tests, not out-of-sample validation.

↔ Report: Methodology, Verification and Validation Approach; Results, V&V Results.

Limitations and scope

Q. Aisle congestion is not modelled. Doesn't that invalidate the results?

It makes the absolute lead times slightly optimistic under heavy load, but all six policies share the assumption symmetrically, so the comparison stays fair. If anything, adding congestion would widen Line-aware's lead, because the baselines saturate the reach-truck fleet far more, up to 44% utilization for the travel-distance policy.

↔ Report: Appendix, Model Assumption Register; Conclusions, Limitations.

Q. How real is the "Actual SAP" baseline?

It is an honestly-labelled SAP-plus-heuristic hybrid. 750 materials sit at their exact decoded SAP rack-bay-position, 2,872 are Kardex-routed and policy-invariant, and the rest fall to a frequency heuristic. 386 multi-bin materials are not split into forward and reserve. We never claim a pure replay; it is the best reconstruction the decoded SAP data allows, and it is our validation ground truth.

↔ Report: Methodology, Data Collection; Conclusions, Limitations.

Q. Throughput barely changes and RT utilization is only 1.3%. Is that believable?

Yes. The system is demand-limited: every policy sees the same arrival stream, so throughput is about 398 a day for all of them, with an ANOVA F near 0.01. The win is in lead and waiting time, not output. The 1.3% reach-truck figure is picking workload only; replenishment is excluded for all six policies equally, so the comparison is fair, and we flag that the live total would be higher.

↔ Report: Results, Policy Comparison and Discussion; Conclusions, Limitations (replenishment caveat).

Q. What about the milk-run trains?

The model represents seven milk-run trains as an available delivery mode, but in the reported runs the operator carries the finished kit to the line, so lead time is attributed to operators and reach trucks only. Enabling milk-run would offload some operator walking; it does not touch the reach-truck queue that drives the result.

↔ Report: Appendix, Model Assumption Register.

Layout and reproducibility

Q. Is the warehouse layout real or invented?

Reconstructed from the real AutoCAD drawing and eleven per-rack pallet PDFs, cross-checked by a blind CAD extraction. Eleven rack rows, A to J and U; 3,101 modelled pallet positions reconcile with the 3,203 in the drawing, the 102-gap being narrower feeder bays plus a cart-parking slot.

↔ Report: Methodology, System Description and Geometry; the top-down layout figure.

Q. Can someone reproduce your results?

Yes. The code is fixed-seed and deterministic given a seed; six policies times twenty replications is 120 runs, exported as a tidy table for Minitab. The software versions and reproducibility notes are in the appendix.

↪ Report: Appendix, Software and Reproducibility.

One-line fallbacks if you blank: “It is distribution-driven, so I can re-slot and re-run.” • “Chi-square is significant by design; read Cramer’s V, 0.22.” • “1.3% RT is picking only, and it is fair across all policies.” • “Sensitivity at $\pm 20\%$ keeps Line-aware first.”